Year 10 Mathematics

Graphs 2

8.5 Mid-points

Mid-points

STARTER · 5 MIN

Date: _____ **Title:** Mid-points

LEARNING OBJECTIVES

- Find the mid-point of a line segment
- Find mid-points with negative and decimal coordinates
- Find an end point given the mid-point

STARTER *Answer in your book. No calculators.*

1. Work out $(4 + 10) \div 2$
2. Work out $(-3 + 7) \div 2$
3. Work out $(-6 + -2) \div 2$
4. Solve $(x + 4) \div 2 = 5$

Starter — Answers

STARTER

Tick or correct in green pen

1. $14 \div 2 = \mathbf{7}$
2. $4 \div 2 = \mathbf{2}$
3. $-8 \div 2 = \mathbf{-4}$
4. $x + 4 = 10, \quad x = \mathbf{6}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the mid-point of a line segment	☐
SILVER	I can find mid-points with negative and decimal coordinates	☐
GOLD	I can find an end point given the mid-point	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Mid-points

INTRODUCTION · 3 MIN

The mid-point is exactly halfway between two points.

x-coordinate

Add the x-coordinates and halve.

$$\frac{x_1 + x_2}{2}$$

y-coordinate

Add the y-coordinates and halve.

$$\frac{y_1 + y_2}{2}$$

Formula

The mid-point is the mean of the coordinates.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Check

Plot the points to see if it looks right.

M is on the line, halfway

Examples 1 and 2 — Finding a Mid-point

EXAMPLES · 5 MIN

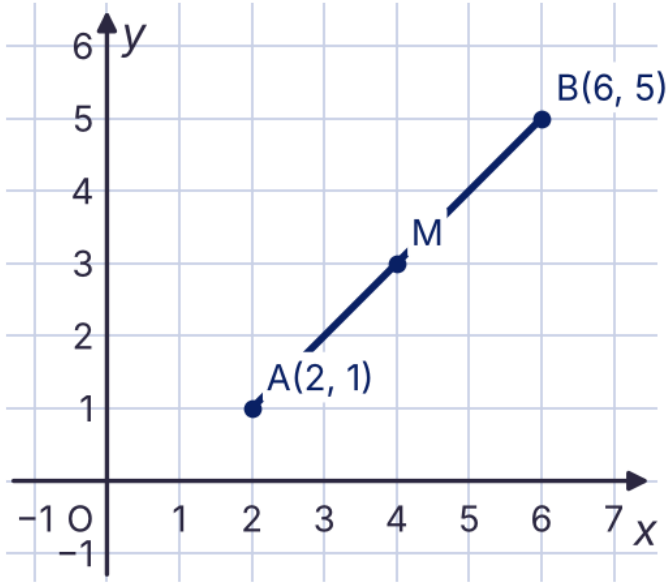
BRONZE

Watch and copy into your book

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Example 1

Find the mid-point of A(2, 1) and B(6, 5).



Example 2

Find the mid-point of P(1, 8) and Q(7, 2).

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$x = \frac{2+6}{2} = 4$$

$$y = \frac{1+5}{2} = 3$$

M = (4, 3)

Tip: add, then halve.

Example 2

$$x = \frac{1+7}{2} = 4$$

$$y = \frac{8+2}{2} = 5$$

M = (4, 5)

Tip: do x and y separately.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the mid-point of each pair of points

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

1. (2, 4) and (8, 10)
2. (1, 3) and (5, 7)
3. (0, 6) and (4, 0)
4. (3, 9) and (7, 1)
5. (2, 5) and (5, 9)

★ (10, 2) and (4, 12)

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. M = **(5, 7)**

2. M = **(3, 5)**

3. M = **(2, 3)**

4. M = **(5, 5)**

5. M = **(3.5, 7)**

★ M = **(7, 7)**

Examples 3 and 4 — Negative and Decimal Coordinates

EXAMPLES · 5 MIN

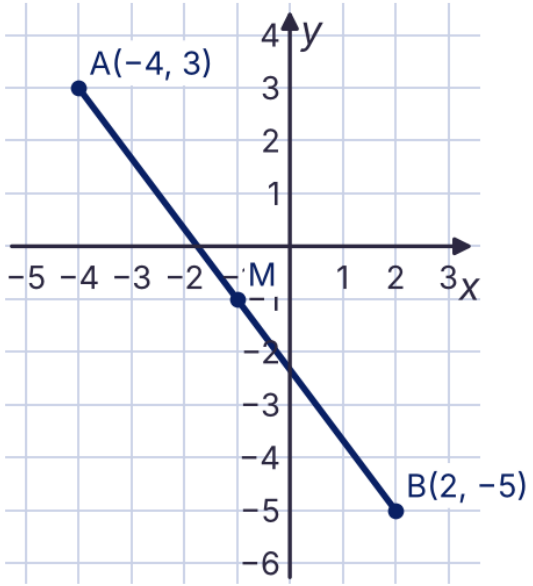
SILVER

Watch and copy into your book

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Example 3

Find the mid-point of A(−4, 3) and B(2, −5).



Example 4

Find the mid-point of C(−3, −2) and D(4, 7).

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x = \frac{-4 + 2}{2} = -1$$

$$y = \frac{3 + (-5)}{2} = -1$$

M = (−1, −1)

Tip: take care with negative signs.

Example 4

$$x = \frac{-3 + 4}{2} = 0.5$$

$$y = \frac{-2 + 7}{2} = 2.5$$

M = (0.5, 2.5)

Tip: the answer can be a decimal.

Task 2

PRACTICE · 9 MIN

SILVER

Find the mid-point of each pair of points

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

1. (−2, 4) and (6, 8)
2. (−5, −1) and (3, −7)
3. (−6, 2) and (−2, −4)
4. (1, −3) and (4, 6)
5. (−7, 5) and (2, −2)

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. M = **(2, 6)**

2. M = **(−1, −4)**

3. M = **(−4, −1)**

4. M = **(2.5, 1.5)**

5. M = **(−2.5, 1.5)**

Examples 5 and 6 — Finding an End Point

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Example 5

(3 marks)

M(5, 4) is the mid-point of AB.

A is (2, 1). Find the coordinates of B.

Example 6

(3 marks)

M(−1, 3) is the mid-point of PQ.

P is (4, 6). Find the coordinates of Q.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$\frac{2+x}{2} = 5$$

$$x = 8$$

$$\frac{1+y}{2} = 4$$

$$y = 7$$

B = (8, 7)

Tip: set up an equation for x and for y.

Example 6

$$\frac{4+x}{2} = -1$$

$$x = -6$$

$$\frac{6+y}{2} = 3$$

$$y = 0$$

Q = (−6, 0)

Tip: quick method: the step from P to M equals the step from M to Q.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

1. M(4, 6) is the mid-point of AB. A is (1, 2). Find B. (3 marks)

2. M(0, 3) is the mid-point of PQ. P is (−4, 7). Find Q. (3 marks)

3. M(2.5, −1) is the mid-point of CD. C is (6, 3). Find D. (3 marks)

4. The mid-point of (k, 4) and (8, 10) is (6, 7). Find k. (2 marks)

5. A circle has diameter AB with A(−3, 2) and B(5, 8). Find the centre. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $x = 7, \quad y = 10, \quad B = \mathbf{(7,\ 10)}$

2. $x = 4, \quad y = -1, \quad Q = \mathbf{(4,\ -1)}$

3. $x = -1, \quad y = -5, \quad D = \mathbf{(-1,\ -5)}$

4. $k + 8 = 12, \quad k = \mathbf{4}$

5. Centre $= \mathbf{(1,\ 5)}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the mid-point of (4, 2) and (10, 8)

SILVER Find the mid-point of (−3, 5) and (1, −1)

GOLD M(3, 2) is the mid-point of AB, A(0, −2). Find B.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE M = **(7, 5)**

SILVER M = **(−1, 2)**

GOLD B = **(6, 6)**

Exam Practice — Finding a Mid-point

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

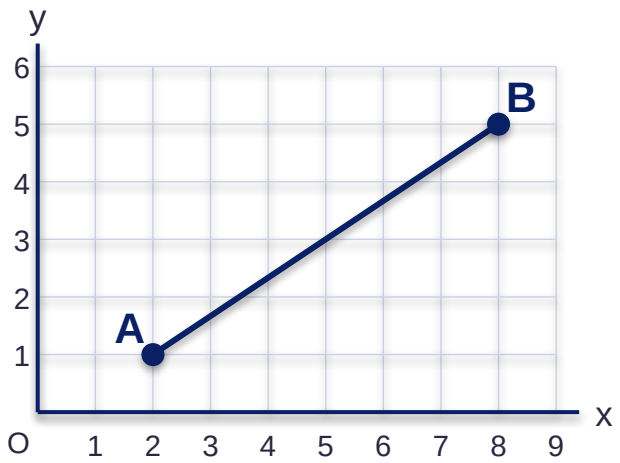
Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Question 1

(2 marks)

A is (2, 1) and B is (8, 5).

Find the coordinates of the mid-point of AB.



(..... , ...)

Question 2

(2 marks)

P is (1, 8) and Q is (7, 2).

Find the coordinates of the mid-point of PQ.

(..... , ...)

Exam Practice — Negative and Decimal Coordinates

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

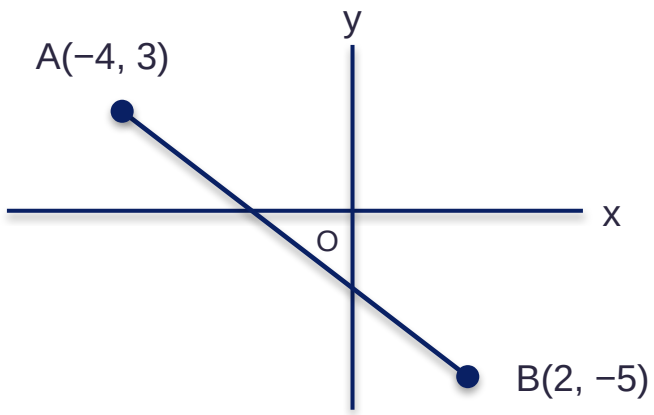
Question 3

(2 marks)

A is (−4, 3) and B is (2, −5).

Find the coordinates of the mid-point of AB.

Diagram NOT accurately drawn



(..... , ...)

Question 4

(2 marks)

C is (−3, −2) and D is (4, 7).

Find the coordinates of the mid-point of CD.

(..... , ...)

Exam Practice — Finding an End Point

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Question 5

(3 marks)

M(3, −2) is the mid-point of AB.

A is (−1, 4).

Find the coordinates of B.

(..... , ...)

Question 6

(3 marks)

The mid-point of (a, 3) and (7, b) is (4, 5).

Find the value of a and the value of b.

a = ... b = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

((2 + 8) ÷ 2, (1 + 5) ÷ 2) **M1**

(5, 3) **A1**

Tip: add and halve each coordinate.

Question 2

((1 + 7) ÷ 2, (8 + 2) ÷ 2) **M1**

(4, 5) **A1**

Tip: the mid-point is the mean of the end points.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

((−4 + 2) ÷ 2, (3 + −5) ÷ 2) — M1

(−1, −1) — A1

Tip: take care adding negatives.

Question 4

((−3 + 4) ÷ 2, (−2 + 7) ÷ 2) — M1

(0.5, 2.5) — A1

Tip: mid-points can be decimals.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

x: −1 → 3 is +4, so 3 + 4 = 7	M1
y: 4 → −2 is −6, so −2 − 6 = −8	M1
B(7, −8)	A1

Tip: apply the same step again from M.

Question 6

(a + 7) ÷ 2 = 4	M1
a = 1	A1
(3 + b) ÷ 2 = 5, b = 7	B1

Tip: form an equation for each coordinate.